ARP, ICMP, DNS

DNS two question : name lookup and caching why important then replication, icmp: two question (how to trace a route and path mtu discovery, what is mtu)

Address Resolution Protocol (ARP):- ARP associates an IP address with the physical address.

Logical Address $\xrightarrow[\text{(ARP)}]{\text{Mapping}}$ Physical Address

[IMP:- As 'IP' uses the Service of Data link Layer, it needs to Know the physical address of the next Hop → ARP]

Mapping of IP Address into MAC Address:-

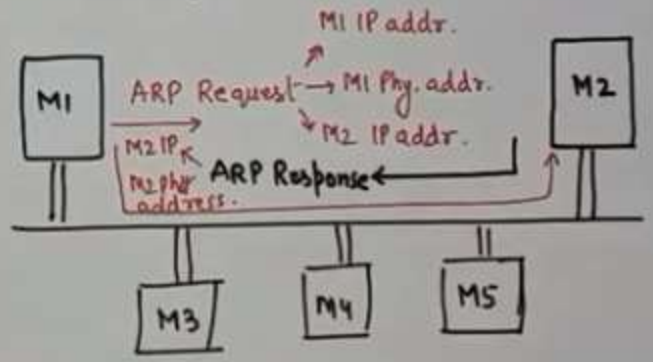
Static Mapping

Table is Created with logical & Physical addr.

Dynamic Mapping → ARP, RARP.

Each time a m/c Knows the logical addr. of another m/c, it Can use a protocol to find Physical Addr.

ARP Protocol:- ARP accepts a logical address from the IP protocol, maps the address to the Corresponding physical address and pass it to the data link Layer.



ARP Request → BroadCast

ARP Response → Unicast

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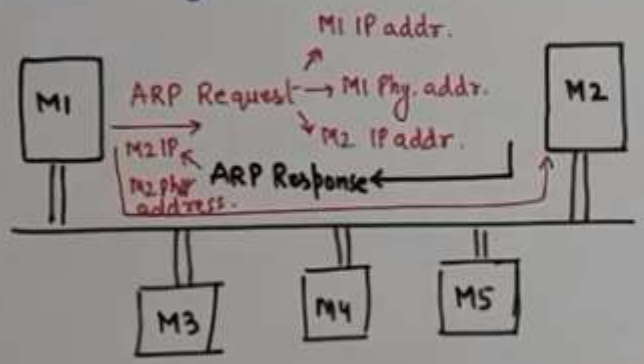

ARP Request → BroadCast

ARP Response → Unicast

ARP operation:- Anytime a HOST or a ROUTER needs to find a physical address of another HOST or ROUTER on its N/w, it sends an ARP Query packet. The packet includes the physical and IP address of the sender and the I/P address of the receiver. Query is broadcast to the N/w, when the intended receipient recognizes its IP address, it sends back an ARP response packet which contains both IP and physical address.

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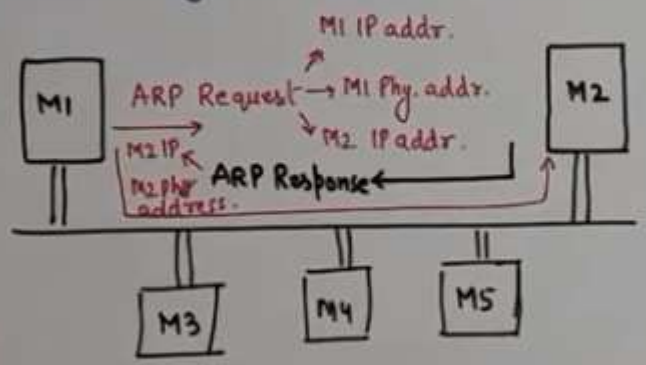

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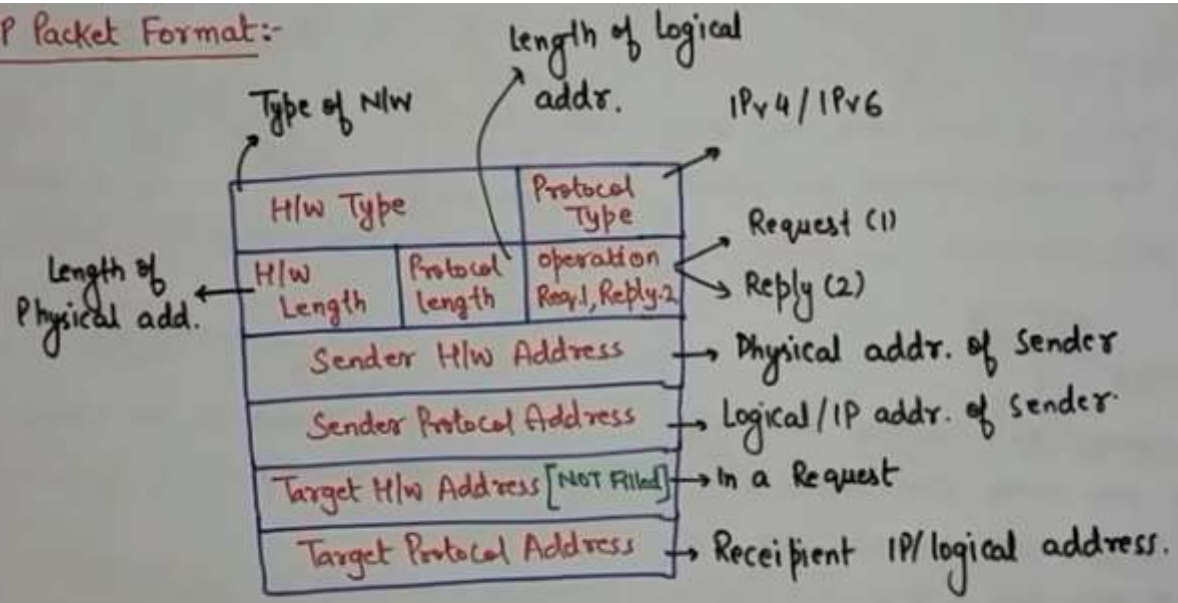
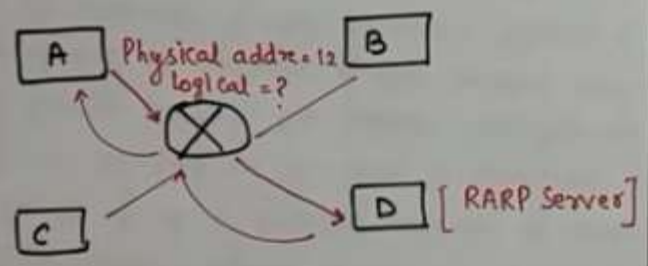
ARP Packet Format:-



Reverse Address Resolution Protocol (RARP):-

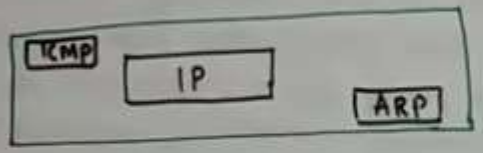
'RARP' Maps a physical address to a logical address.

Physical Address $\xrightarrow{\text{(Mapping)}}$ Logical/IP Addr.

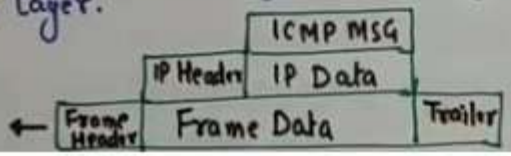


Internet Control Message Protocol (ICMP) :- ICMP is designed to overcome the following two problems (N/W) with IP Protocol:-

- (i) No Error Reporting/correcting mechanism.
- (ii) Lacks a mechanism for Queries.



→ ICMP messages are not directly passed to the data link layer. Instead, the messages are first encapsulated inside IP datagrams before going to the lower layer.



ICMP Messages:-

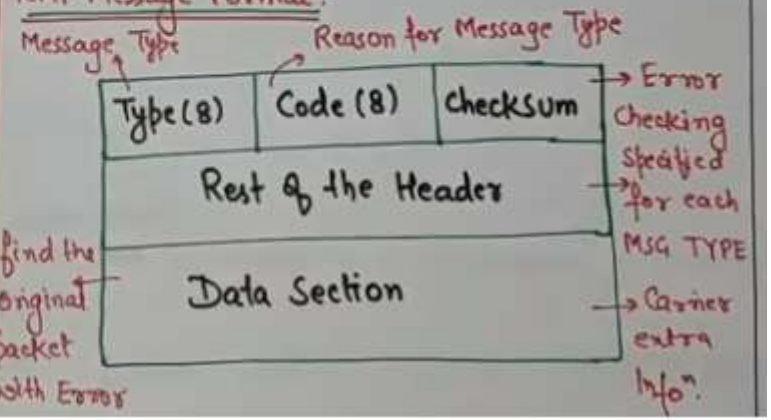
→ **Error Reporting**

Reports Problem that Router or a Host may encounter while Processing of IP Packet.

Query

Fetch Important/ Specific Info^n from a Router or HOST.

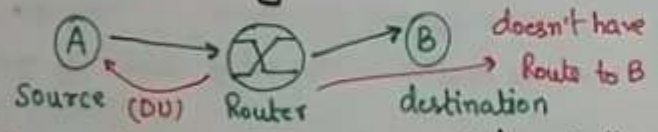
ICMP Message Format:-



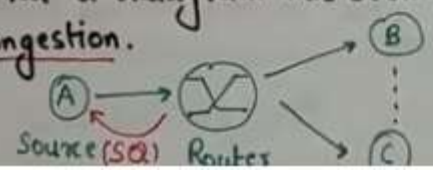
Error Reporting Messages:- ICMP always reports error message to the original Source.

[ICMP does not Correct Errors]

(i) Destination Unreachable:- Whenever a Route to the destination is not found, datagram is discarded and Router/Host sends Destination Unreachable Message back to the Source.



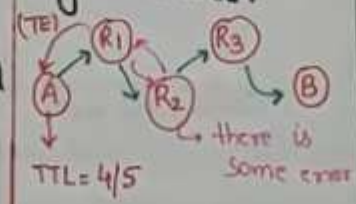
(ii) Source Quench:- This message informs the Source that a datagram has been discarded due to Congestion.



(iii) Time Exceeded:- It is generated in following two cases.

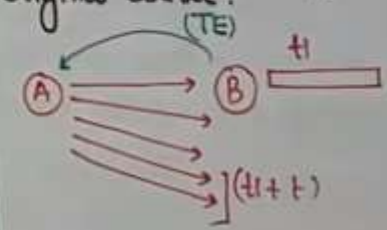
Code-0

Whenever a Router '↓' a datagram with TTL Value to Zero (0), it discards the datagram and sends a time-exceeded Msg to the original Source.



Code-1

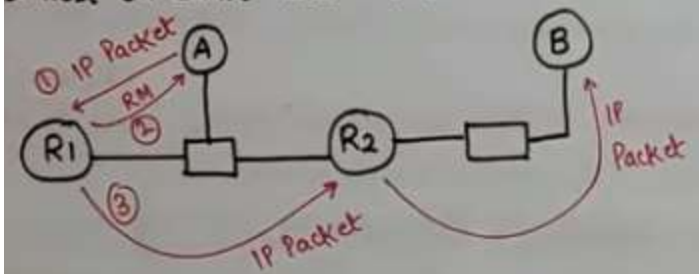
When the final destination doesn't receive all fragments in a set time, it discards the received fragments and sends a time-exceeded Msg. to the original Source.



(iv) Parameter Problem:- If router/host discovers an ambiguous or missing value in any field of the datagram, it discards datagram and sends a Parameter Problem message back to the source. → Code '0' → Error/ Ambiguity in one of the Header field.

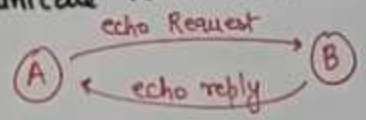
→ Code '1' → Required Part of the option is missing.

(v) Redirection:- It is sent from a router to a host on same local N/W.



ICMP Query Messages:- It is used to diagnose some N/W problems.

(i) Echo Request and Reply:- Designed for the diagnostic purpose. These messages are used to determine whether two systems can talk/ communicate with each other.



(ii) Timestamp Request and Reply:- Two machines uses these messages to determine the Round-Trip time needed for an IP datagram to travel b/w them. → Sending Time = Receive - Original

→ Receiving Time = Returned - Transmit

→ Round-Trip = Sending + Receiving.

ICMP

Q: (1) What do you mean by Internet Control Protocol?

Ans: **Internet Control Protocol:**

In addition to IP datagram which is used for data, internet has some other control protocols like ICMP, ARP, RART, DHCP and BOOTP. These protocol are known as Internet control protocols.

Q: (2) Discuss: Using ICMP messages to Test Reachability. (2003)

Ans: **Using ICMP Messages to Test Reachability:**

Ping uses the ICMP echo request and echo reply messages. When invoked, ping sends an IP datagram that contains an ICMP echo request message to the specified destination. After sending the request, it waits a short time for the reply.

If no reply arrives, ping retransmits the request.

If no reply arrives for the retransmissions, ping declares that the remote machine is not reachable.

2008 (Mark - 15)

Q: (3) Discuss the ICMP message for finding the sequence of routers & determining the MTU Path.

Ans:

(4) What are the uses of ICMP messages for (trace route and path MTU Discovery)?

(Mark-20)

Using ICMP To Trace A Route:

ICMP messages are used by the traceroute tool when it constructs a list of all routers along a path to a given destination. Traceroute simply sends a series of data-grams & waits for a response to each.

Traceroute sets the TIME TO LIVE value in the first datagram to 1 before sending the datagram. The first router that receives the datagram decrements the time to live, discard the datagram & sends back an ICMP time exceeded message.

After it discover the address of the first router, traceroute sends a datagram with TIME TO LIVE set to 2.

The first router decrements the counter and forwards the datagram; the second router discards the datagram & sends an error message. Similarly, once it has received an error message from a router that is distance 2, traceroute sends a datagram with TIME TO LIVE set to 3, then 4, and so on.

Using ICMP for path MTU discovery:

(Maximum Transmission Unit)

Technically, the smallest MTU along a path from a source to a destination is known as the path MTU.

Using ICMP a host can determine the path MTU. To determine the path MTU, IP software on a host sends a sequence of probes, where each probe consists of a datagram that has the header bit set to prevent fragmentation. If a datagram is larger than the MTU of a network along the path, the router connected to the network will discard the datagram and send the appropriate ICMP message to the host. The host can then send a smaller probe until one succeeds. As with traceroute, a host must be prepared to retransmit probes for which no response is received.

DNS (Domain Name System)

- **Optimizations of DNS Performance:**
- Two primary optimizations:
  (i) Replication & (ii) Caching.
- DNS caching is more important than replication because caching affects most of the system.

Question

- DNS কিভাবে Name Resolve করে। (V.V.I) (100%)

Ans: Although the names used by the DNS is very convenient to humans, it can't be used for communication through the internet. The translation of a domain name into an equivalent IP address is called name resolution & the name is said to be resolved an address. Software to perform the translation is known as name resolver software.

The resolver sends a UDP packet to a local DNS server, which searches for the name in its database. If the name is found, it returns the IP address to the resolver, which in turn informs it to the client. After having the IP address, the client then establishes a TCP connection with a destination node.

Q: How a resolver looks up a remote name in eight steps?

A: To map a name onto an IP address, an application calls a library procedure called the resolver, passing the name as parameter.

Step 1: A query is sent to local server. The query ~~step~~ contains domain name sought, type and class.

Step 2: If the local name server never had a query for this domain before and knows nothing about it, it will ask few other name servers. If none of them knows, it will send UDP packets to the server

Step 3: A server must know all of it's children so, request is forwarded with specific name.

Step 4: In turn, request should be made with name which must have authoritative resource records.

Since each request from a client to server in steps 1 to 4 the resource record that was requested works its way back in similliar 4 more steps.

Q: Why DNS caching is more important than replication? / Optimization of DNS?

A: Caching affects most of the system. When server looks up a new name, it replaces a binding in its cache. Before contracting ~~new~~ another server to request a binding, it checks its cache. If the cache contains the answer, server uses the cached answer to generate a reply. So it saves times and enables more efficient system.

In replication if a new site is added in internet, the site configures its local

DNS server with a list of root servers. A site of europe will choose a root server in europe.

Q:- Define SMTP, POP, HTTP, POP, DHCP, SMB, DSL, ADSL, ATM, ISDN, SONET.

How DHCP server dynamically assigns IP address to a host?
DHCP is an abbreviation for Dynamic Host Configuration Protocol. It is an application layer protocol used by hosts for obtaining network setup information. The DHCP is controlled by DHCP server that **dynamically distributes** network configuration parameters such as IP addresses, subnet mask and gateway address.

What is Dynamic host configuration protocol?

Dynamic – Automatically
Host – Any computer that is connected to the network
Configuration – To configure a host means to provide network information(ip address,subnet mask,Gateway address) to a host
Protocol – Set of rules
Summing up, a DHCP server dynamically configures a host in a network.

Disadvantage of manually Configuring the host: Configuring a host when it is connected to the network can be done either manually i.e., by the network administrator or by the DHCP server. In case of home networks, manual configuration is quite easy. Whereas in the large networks, the network administrator might face many problems. Also, the manual configuration is prone to mistakes. Say a Network administrator might assign an IP address which was already assigned. Thus, causing difficulty for both administrator as well as neighbors on network.

Configuring a host using DHCP

To configure a host, we require the following things:

Leased IP address – IP address to a host which lasts for a particular duration which goes for a few hours,few days or few weeks.

Subnet Mask – The host can know on which network it is on.

Gateway address – The Gateway is the Internet Service Provider that connects user to the internet.The Gateway address lets the host know where the gateway is to connect to the internet.

DHCP Entities

DHCP server: It automatically provides network information(IPaddress, subnet mask, gateway address) on lease. Once the duration is expired, that network information can be assigned to other machine. It also maintains the data storage which stores the available IP addresses.

DHCP client: Any node which request an IP address allocation to a network is considered as DHCP client.

DHCP Relay Agent: In case, we have only one DHCP server for multiple LAN's then this Agent which presents in every network forwards the DHCP request to DHCP server. So, using DHCP Relay Agent we can configure multiple LAN's with single server.

How DHCP server assigns IP address to a host?

DHCPDISCOVER: When a new node is connected to the network, it broadcasts the DHCPDISCOVER message which contains the source address as 0.0.0.0 to every node on the network including server. DHCP server on receiving the message, returns the DHCPOFFER message to the requested host which contains the server address and new IP address to the node.

DHCPOFFER: If there are multiple servers on the network, host receives multiple DHCPOFFER messages. It is up to the host to select a particular message.

DHCPREQUEST: The requested host on receiving the offer message, it again broadcasts the DHCPREQUEST message on the network with the address of the server whose offer message is accepted by the host. The server which pertains to that server address sent by the host checks whether the address to be assigned to the node is available in the data storage.

DHCPACK : If the address is assigned , it marks the IP address in the storage as unavailable to ensure consistency. Now, the server sends DHCPACK packet to the requested host which contains network information(IP address, subnet mask, gateway address). In case, if the address is assigned to other machine meanwhile, then the server sends the packet DHCPNAK to the requested host indicating that the IP address is assigned to some other machine.

DHCPRELEASE : And finally, If the host wants to move to other network or if it has finished its work, it sends the DHCPRELEASE packet to the server indicating that it wants to disconnect. Then the server marks the IP address as available in the storage so that it can be assigned to other machine.